

Practical Assignment - 5

Department: Computer Engineering & Applications

Course: BCA

Subject: Database Management System Lab (BCAC0816)

Year: 1ST

Semester: 2ND



Create these tables which consist of following attributes

College

Column Name	Data type	Size
cName	varchar2	10
cstate	varchar2	10
enrollment	int	

Student

Column Name	Data type	Size
sID	int	
sName	varchar2	10
CGPA	number	2,1
marks	int	
DoB	date	

Apply

Column Name	Data type	Size
sID	int	
cName	varchar2	10
major	varchar2	20
decision	char	1

Insert the following data to these tables:

Student

sID	sName	CGPA	Marks
123	Amit	3.9	1000
234	Balbir	3.6	1500
345	chirag	3.5	500
456	Dev	3.9	1000
567	Eshan	2.9	2000
678	Faizal	3.8	200
789	Garvit	3.4	800
987	Himani	3.7	800
876	Ishan	3.9	400
765	Jay	2.9	1500
654	Amit	3.9	1000
543	chirag	3.4	2000

College

cName	state	enrollment
Stanford	CA	15000
Berkeley	CA	36000
MIT	MA	10000
Cornell	NY	21000
Harvard	MA	50040

Apply

sID	cName	major	decision
123	Stanford	CS	Y
123	Stanford	EE	N
123	Berkeley	CS	Y
123	Cornell	EE	Y
234	Berkeley	biology	N
345	MIT	bioengineering	Y
345	Cornell	bioengineering	N
345	Cornell	CS	Y
345	Cornell	EE	N
678	Stanford	history	Y
987	Stanford	CS	Y
987	Berkeley	CS	Y
876	Stanford	CS	N
876	MIT	biology	Y
876	MIT	marine biology	N
765	Stanford	history	Y
765	Cornell	history	N
765	Cornell	psychology	Y
543	MIT	CS	N

State the SQL Queries for each of the following:

- Q1. Count the total number of Students.
- Q2. Calculate the average CGPA of all Student.
- Q3. Determine the minimum and maximum GPA. Rename the titles as 'max_CGPA' and 'min_CGPA' respectively.
- Q4. Count the number of students having CGPA greater than or equal to 3.7.
- Q5. Find Maximum, Average, Minimum, total CGPA of all student.
- Q6. Find total number of colleges in our Application Database.
- Q7. Find how many different majors student had applied in.
- Q8. Find total no. of Applications in our Application System's Database.
- Q9. Find average of all distinct CGPA.
- Q10. Display the total number of application accepted.
- Q11. Find number of students having CGPA>3.4 and coming from high school having marks>1000.
- Q12. Find how many students applied to 'marine biology'.
- Q13. Find how many applications were rejected and accepted by the colleges.
- Q14. Give state and number of colleges of a state that has more than 1 college.
- Q15. Find the name of students that are duplicate.